Astrarium — User Manual

For version 1.2

29 July 2026

Contents

1. Introduction

Astrarium is a planisphere and astronomical simulator for iPhone and iPad. It shows the sky as it is now, but it also lets you drive the clock freely — back into the past, forward into the future — and it puts planetary phases, eclipses, meteor showers, satellites and comets on the same chart.

Positions come from the JPL DE421 ephemeris, the IAU precession and nutation models and IERS Earth-orientation data, and are accurate enough to plan real observing sessions with.

1.1 Requirements

Devices	iPhone / iPad (iOS and iPadOS 17.0 or later)
Orientation	Portrait and landscape both supported
Price	Free. No in-app purchases, no advertising
Network	The chart and all the astronomy run on the device — it works offline
Data collection	None. Your site and settings stay on the device

An internet connection is needed only to **update** three things:

- satellite orbital elements (TLEs),
- comet and asteroid elements (from the MPC and JPL),
- your current position (this uses location services, not the network). Where Precise Location is switched off, the app asks for it once, for that single fix: an approximate position can be kilometres out, and that changes what is visible near the horizon.

1.2 First steps

1. On launch, the app shows the sky over Tokyo at the present moment.
2. Press 📍 in the top bar and allow the use of your location. Astrarium switches to your position and tells you the nearest registered site and how far away it is. If you would rather not share your location, pick a site from the list or type in coordinates (chapter 3).
3. Outdoors, press 🔴 (night vision). The whole interface turns red so that it does not spoil your dark adaptation.

2. The screen

The screen is divided into five bands, from top to bottom.

Band	Where	What it does
Top bar	Top	Clock, observing site, location, panel, language, night vision
Chart	Middle	The sky itself — driven with your fingers
Quick bar	Below the chart	Magnitude slider and twelve common toggles
Time bar	Below that	View mode, grids, moving and playing the clock
Signature	Bottom	Logo and author, both links

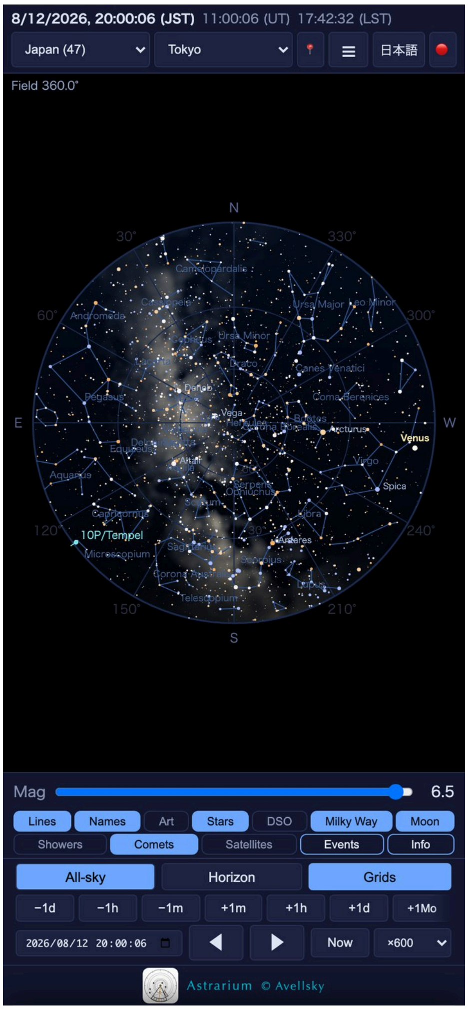

Figure 1. The all-sky view (Tokyo, 12 August 2026, 20:00)

2.1 The top bar

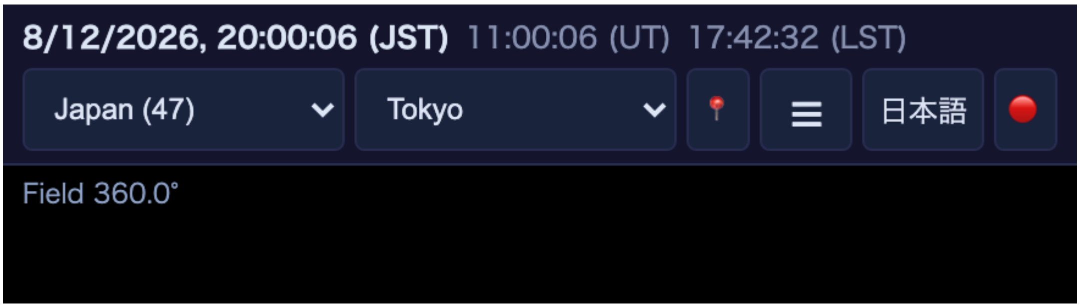

Figure 2. The top bar: site category and site, location, panel, language, night vision

- **Clock** — local time at the observing site, Universal Time (UT) and Local Sidereal Time (LST) side by side. LST tells you which right ascension is on the meridian right now, which is what you need when pointing a telescope.
- **Site category** — Japan's prefectural capitals, Japanese dark-sky sites, Japanese observatories, the world's observatories, major cities other than capitals, the capitals of each continent, and the Antarctic research stations. Each menu is named for exactly what is in it: "Asia · capitals" holds one national capital per country. Choosing observatories adds a second **Region** menu so you can narrow the list down by continent.
- **Site** — the place itself. The prefectures and the dark-sky sites run north to south; the world cities are alphabetical, by the reading of the name when the app is in Japanese.
- 📍 — use your current position.
- ≡ — open or close the detail panel (six tabs).
- **EN / 日本語** — switch language. Object names, constellation names and every description change with it.
- 🔴 — night vision; press again to return to normal colours.

2.2 The signature line

The logo at the bottom opens the Astrarium home page and `© Avellsky` opens the author's page, both in the browser.

3. Choosing where you are

There are three ways to set the observing site.

Pick it from the list

Use the category and site menus in the top bar. The app carries all 47 the capital of each of the 47 Japanese prefectures — "Sapporo (Hokkaido)", named with its prefecture and listed north to south — recommended dark-sky sites, every national capital named with its country, the major cities that are not capitals, the Antarctic research stations, and the world's observatories with their MPC codes. Each place appears in one menu only.

Use your current position

Press 📍 and choose "While Using the App" when iOS asks. Your latitude and longitude become the observing site, and the nearest registered site is shown for reference.

Type the coordinates in

≡ → **Info** tab → "Enter coordinates manually". Fill in latitude, longitude and elevation and press "Use this position". "Fill from current location" copies your present position into the fields.

Changing the site updates the clock, the rise and set times and the whole geometry of the chart together.

4. Driving the chart

4.1 All-sky and horizon views

Switch with **All-sky** and **Horizon** at the left of the time bar.

- **All-sky** — the whole sky as a circle centred on the zenith, north up and east to the left, exactly as on a cardboard planisphere. The Milky Way is drawn in this view.
- **Horizon** — part of the sky drawn as you would actually see it looking out. The menu beside the button chooses which way you face: **N**, **E**, **S**, **W**.

Figure 3. Horizon view, facing south, 110-degree field

4.2 Gestures

Gesture	All-sky	Horizon
One-finger drag	**Diurnal rotation** — the sky turns about the celestial pole and the clock follows your finger	Pans the view
Two-finger pinch	Zoom about the point between your fingers	The same
Double tap	Recentre on the zenith	—
Tap an object	Opens its information popup	The same
Mouse wheel	Zoom	Zoom
SHIFT + drag	Pans the view centre without rotating	—

Dragging in all-sky view is the planisphere gesture: the dial turns and the simulated time turns with it, so you can feel out how high Orion will be in two hours' time with a fingertip.

4.3 What a tap tells you

The popup gives the name, right ascension and declination, azimuth and altitude and the magnitude, with buttons underneath. **The readings follow the clock while the card is open.** Azimuth and altitude are recomputed twice a second, α and δ, distance, elongation and phase angle each time the sky is refreshed, and the rise, transit and set times when the local date turns over. "(below the horizon)" is added once the object has set.

The Sun, the Moon, the planets and the moons drawn in the zoom view also carry a sentence or two on what they are.

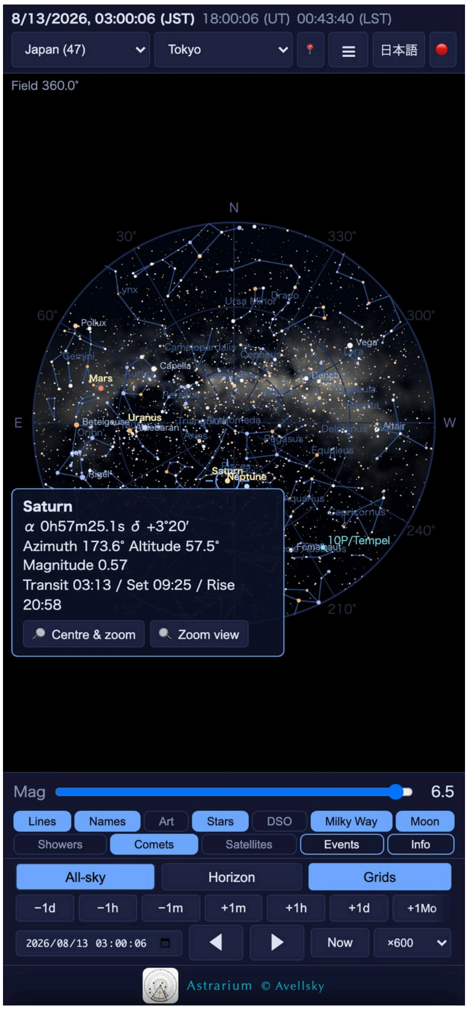

Figure 4. Saturn tapped: coordinates, magnitude, rise and set, and two buttons

- 🔎 **Zoom in** — puts the object in the middle of the screen, zooms in and starts **tracking** it. While tracking, the object stays centred as the clock runs. Release it with the **Release** button on the chart or the `Esc` key; panning the view does not break the tracking.
- 🔍 **Details** (Sun, Moon and planets) — opens a window that draws the body as a disc, recomputed every five seconds so its distance, elongation, phase angle and libration follow the clock too. The Moon comes with its age, illuminated fraction, libration and popular name (Pink Moon, supermoon and so on); Jupiter with the four Galilean moons; Saturn with the rings and Titan; Mars with Phobos and Deimos. You can drag inside the window.

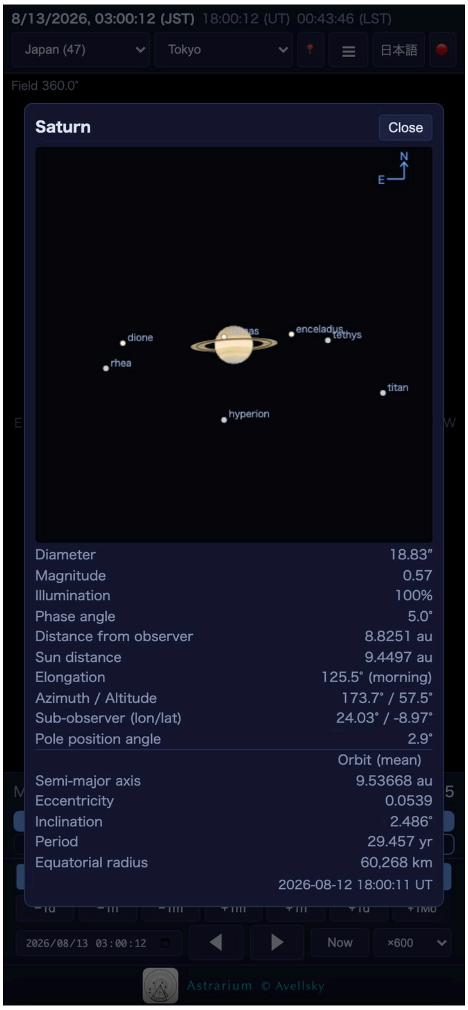

Figure 5. The zoom view of Saturn, with the rings and its moons

Picking a deep-sky object brings up a photograph and a short description; a comet or asteroid brings up its orbital elements and estimated brightness; a satellite brings up its range, height and apparent magnitude with a **Next visible passes** button. Deep-sky names, constellations and descriptions are available in both languages; an object with no Japanese name is listed by its catalogue number in Japanese, with the English name beside it.

Tapping a meteor shower’s radiant gives its azimuth and altitude as well as its coordinates, and these follow the clock — so you can see at a glance whether the radiant is up yet.

With constellation art switched on, tapping one of the twelve zodiacal constellations shows the corresponding astrological sign and its birth dates, together with a note that precession has moved the signs about a month away from where the Sun really is.

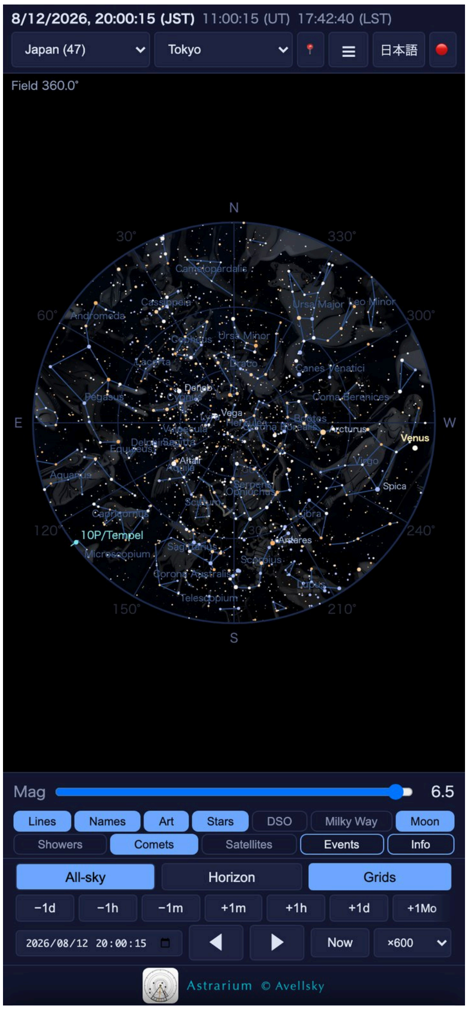

Figure 6. The whole sky with the constellation figures drawn

5. Moving through time

The time bar has two rows.

5.1 Steps and playback

Control	What it does
−1d / −1h / −1m	Back one day, hour or minute
+1m / +1h / +1d	Forward one minute, hour or day
+1Mo	Forward one synodic month (29.53 days) — jumps to the same lunar phase
Date-time field	Go straight to a date and time
◀ / ▶	Play backward / forward (press again to stop)
Now	Return to the present
×1 ··· ×6000	Playback speed; at ×600 one second of yours is ten minutes of sky

With a keyboard, ← and → move ±10 minutes, with `Shift` ±1 day, with `Alt` ±1 minute, and ↑ ↓ move ±1 hour.

5.2 Grids

The **Grids** button opens a menu of overlays; you can have several at once.

Overlay	What it is
Alt-Az grid	Horizontal coordinates — azimuth and altitude
RA-Dec grid	Equatorial coordinates, the system star catalogues use
Ecliptic grid	Ecliptic coordinates; the planets and the Moon keep to this band
Galactic grid	Galactic coordinates; the Milky Way lies along latitude 0°
Celestial equator	Drawn as a single great circle
Ecliptic	The Sun’s path, likewise
Galactic plane	The great circle at galactic latitude 0° — the centre line of the Milky Way
Meridian	The north–south great circle objects transit

The spacing of the lines and the precision of the labels follow the zoom level.

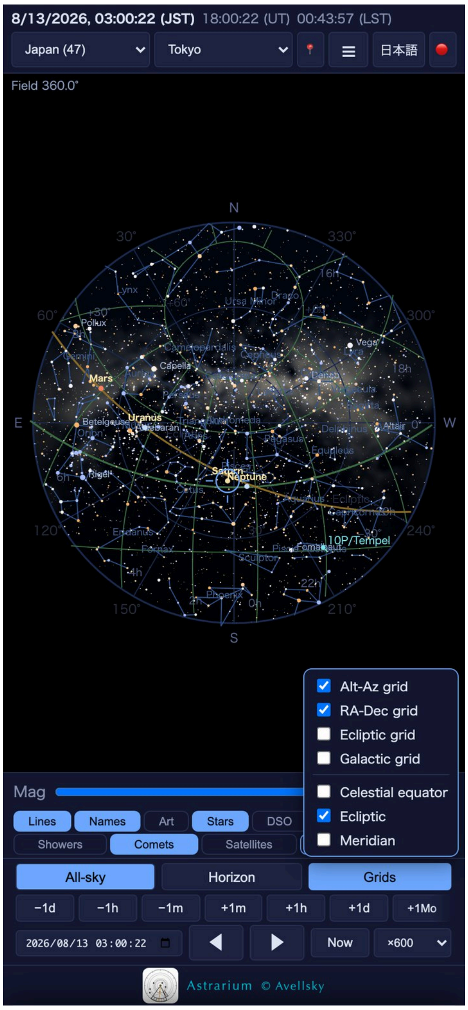

Figure 7. The grid menu, with the RA-Dec grid and the ecliptic drawn

6. What the chart shows

6.1 The quick bar

The slider on the left is the **magnitude limit**. Lower it and the faint stars go, which is much closer to a city sky; raise it and the chart goes to magnitude 6.5, the naked-eye limit.

The first row controls the chart itself, the second the kinds of object.

Button	Shows
Lines	The figures of the 88 constellations
Names	Constellation names
Art	Constellation drawings (86 of them; shown whatever the magnitude limit)
Stars	Names of the named stars
DSO	Messier and Caldwell objects
Milky Way	The Milky Way — drawn in the all-sky view, unbroken down to the horizon
Moon	Sky brightness from moonlight
Showers	Radiants and expected rates of the active meteor showers (chapter 7)
Comets	Every comet currently bright enough (fetches elements the first time)
Satellites	Satellites on the chart (fetches TLEs the first time)
Events	Opens the Events tab
Info	Opens the Info tab

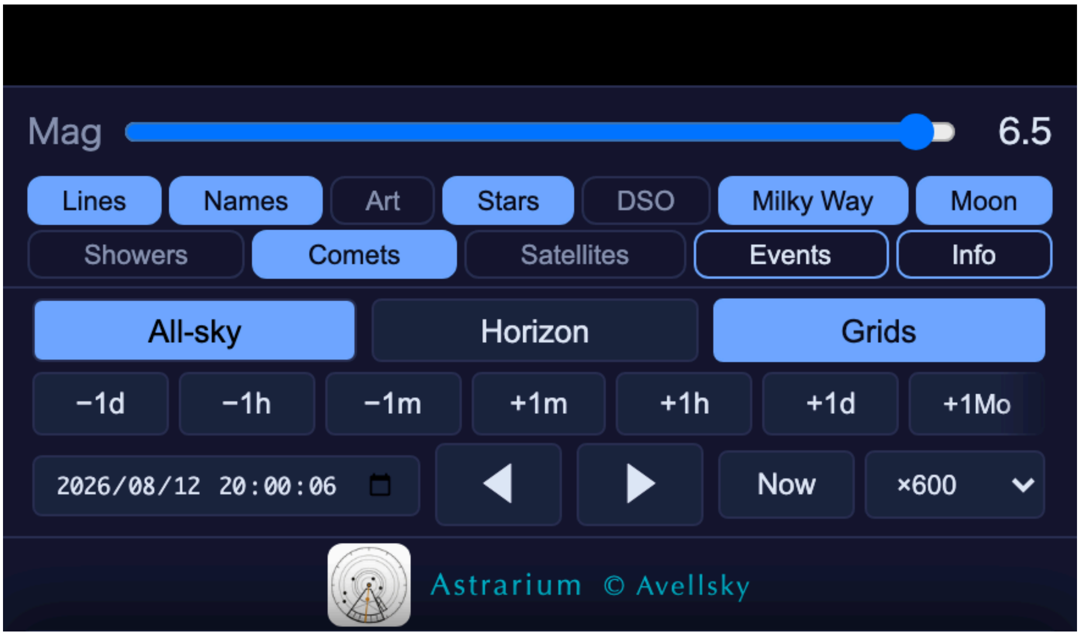

Figure 8. The quick bar and the time bar

6.2 The View tab (≡ → View)

Everything the quick bar does not carry lives here.

- **Planets** — show or hide the planets.
- **Sunlight** — daylight and twilight. Switch it off to draw the sky as if it were always dark, which is how you check where the stars are in the daytime.
- **Mirror** — flips the image for a telescope finder that inverts.
- **Long exposure** — keeps the trails instead of letting them fade, so the tracks build up as they would on film (switching it on switches trails on too).
- **Star trails** — leaves a trail behind moving objects during playback, so diurnal motion and satellite passes are drawn as lines.
- **FOV frames** — rectangles for a camera field or circles for an eyepiece, laid over the chart. “Add frame”, then give it a sensor size and focal length, or an apparent field and a magnification. Drag the centre to move the frame, click the centre to zoom to it, drag a corner to rotate. Useful for framing a photograph, or for seeing what will fit in the field.
- **Mag limit** — the same slider as on the quick bar.

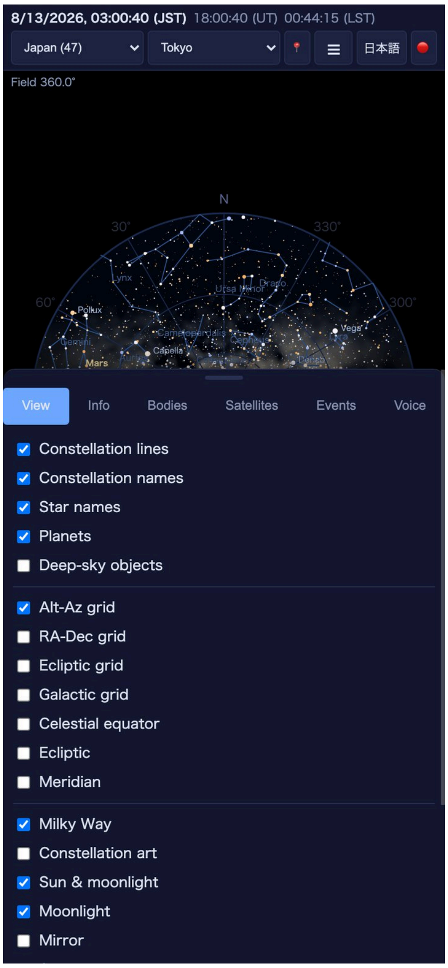

Figure 9. The View tab

6.3 Export as PNG or PDF

Saves the chart for the current time and site as an image. Because these are meant to be printed and taken outside:

- the background is white paper and the stars and lines are black,
- sky brightness (sunlight and moonlight) is left out,
- the Milky Way is drawn **as contours only**, not filled,
- the date, time and `Astrarium © Avellsky` appear in the bottom right corner.

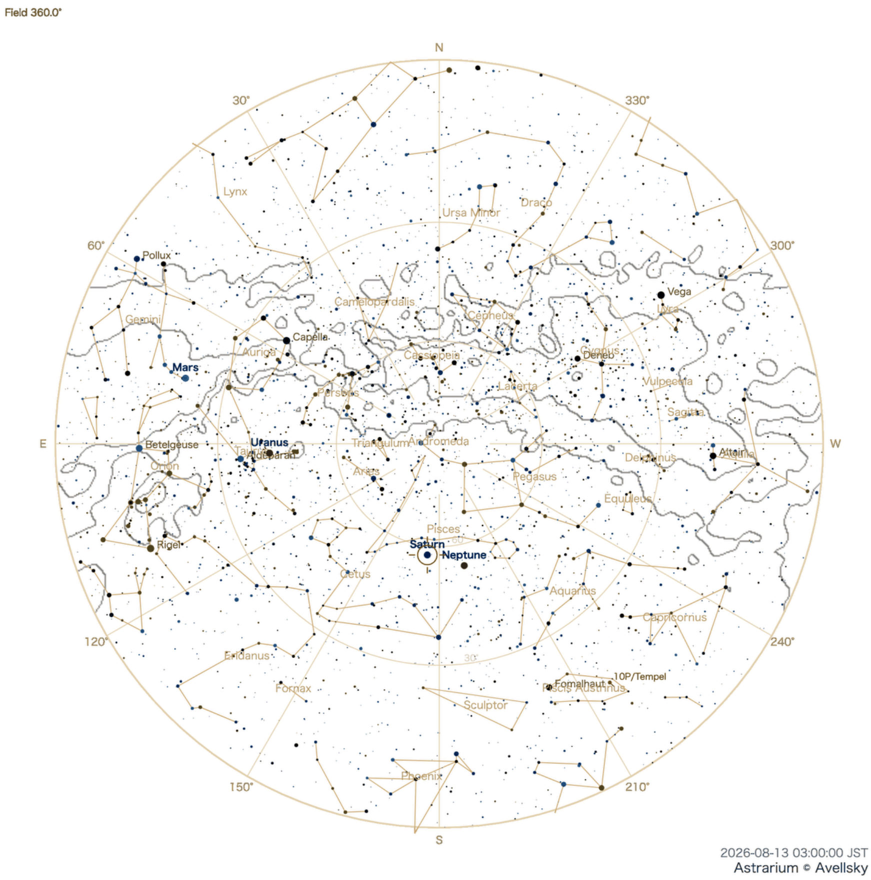

Figure 10. An exported chart: the Milky Way drawn as contours on white paper

6.4 Demo and recording

Demo is a tour of about four minutes. It plays without sound; each scene is described in captions below the chart. It runs through twilight, the Milky Way, a planet, Saturn's rings, FOV frames, the Moon, a meteor shower, an ISS pass, a Starlink train, two Messier objects (M3 and M16, with photograph and description), a simulated ISS flight, a long exposure, the event list, the language switch, the spoken summary and diurnal motion. Touching the screen ends the tour and restores what you had.

It takes a moment to get ready before the first scene appears — a note under the button says so. While it runs, the button turns red and reads **Stop demo**, so you can end it at any point.

Record saves the moving chart as a video (if the device cannot do it, the button says so). Start recording during the demo and the file saves itself when the tour ends.

7. Meteor showers

Use the **Bodies** tab or the **Showers** button on the quick bar.

7.1 The list

The menu holds the principal showers of the year, sorted by **how close their maximum is to the date on the chart**. Choosing one moves the clock to **the moment of that maximum** and opens the details; where the maximum falls in daylight, it moves to about 1 a.m. that night instead. Clicking the radiant on the chart itself leaves the clock where it is.

Field	Meaning
Maximum	Given both in Universal Time and in the local time of your site
Radiant	Right ascension and declination, with its altitude at maximum and now
Entry speed	How fast the meteoroids meet the atmosphere
Population index r	The proportion of bright meteors; the smaller, the brighter
Parent body	The comet or asteroid the stream came from
Expected rate	Meteors per hour, from the model below

7.2 How the rate is worked out

The figure in “per hour” is not the ideal ZHR — it is **what you can expect from your site at that moment**. Four factors are multiplied together:

1. **Distance from the maximum.** A Gaussian profile: about a day either side the rate has fallen to the sporadic background, and outside the activity period nothing is shown at all.
2. **Radiant altitude.** The lower the radiant, the fewer meteors; below the horizon, none.

3. **Twilight.** Zero between sunrise and sunset, reduced during twilight in proportion to how bright the sky is.
4. **Moonlight.** Reduced according to the Moon's phase and altitude.

Active showers are also listed in an orange box at the top of the chart. Tapping a name in the box opens its details; **Release** clears the box (and turns off the Showers button on the quick bar with it).

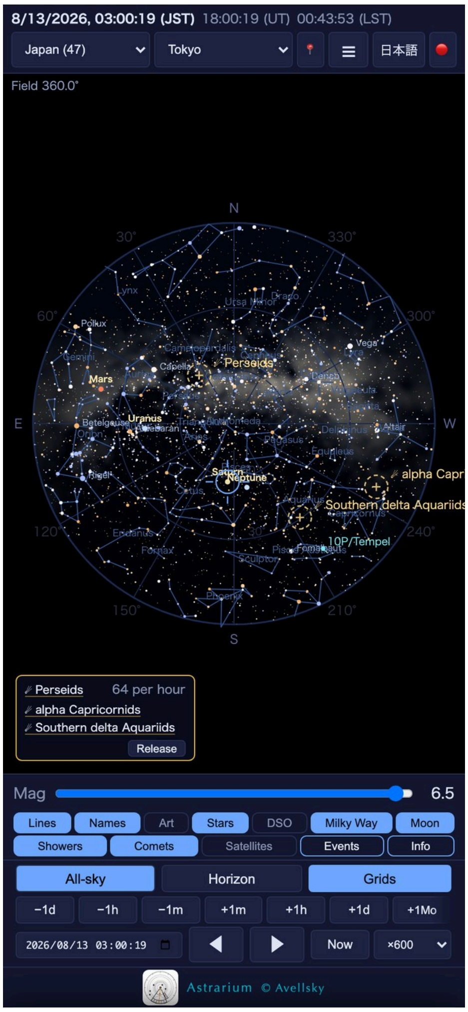

Figure 11. The Perseid maximum (13 August, 3 a.m., 64 per hour expected)

8. Comets and asteroids

The lower half of the **Bodies** tab.

- **List bright now** — the comets and asteroids in the sky, brightest first.
- **Search** — by any part of a name or designation. If the object is not in the local data, Astrarium asks JPL for it and adds it (needs a connection).
- **All / Comets / Asteroids** — filters the list.
- **Selected** — the objects drawn on the chart. Anything you put here is plotted.

- **Update from MPC** — fresh comet elements from the Minor Planet Center and asteroid elements from JPL. The date of the last fetch is shown underneath.

Tapping a comet on the chart gives its distance from the Sun and from the Earth, phase angle, estimated diameter, orbital elements, period and epoch.

9. Satellites

The **Satellites** tab, or the **Satellites** button on the quick bar.

1. Press **Download / update TLEs** first (needs a connection). If the elements were fetched less than two hours ago the app says they are current rather than downloading them again. Several groups are fetched at the same time, so the wait is the slowest one rather than the sum, and the seconds are counted out while it runs. If nothing has answered after 25 seconds — 60 for Starlink — it stops and says so. Starlink is about 1.7 MB and over ten thousand objects, and takes a further ten seconds or so to propagate once it has arrived.

   How often you may fetch. CelesTrak asks that a given group be fetched at most once every two hours — that is how often the elements are regenerated. Requests at shorter intervals are answered with an error, and persisting gets the address blocked altogether. Astrarium records when each group was fetched and, within two hours, answers from what it already has without touching the network. That is not thrift: it is what keeps you from being blocked.

2. Switch on **Show on chart** and the satellites start moving across the sky.
3. Choose which groups to plot.

Group	Contents
Space stations	The ISS and Tiangong
Brightest (visual)	The satellites bright enough to see
Science	Research and observation satellites
Starlink	The Starlink constellation
Weather	Meteorological satellites

- **Mag limit** — 4 by default, which keeps the list to what the eye can see.
- **Sunlit only** — show only satellites out of the Earth’s shadow, that is, the ones actually catching sunlight. These are the only ones you can see.

- Tapping a satellite gives its range, height, apparent magnitude and how old its elements are. **Next visible passes** lists every pass over the next five days.
- Above ×600 playback the satellites are paused, because the propagation cannot keep up.

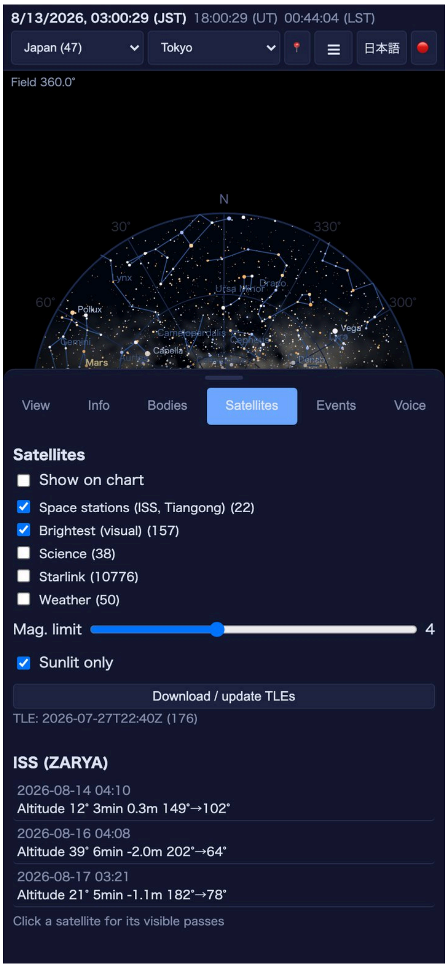

Figure 12. The Satellites tab, with the next passes of the ISS

10. Events

The **Events** tab lists eclipses, oppositions and conjunctions, solstices and equinoxes, meteor maxima, close approaches of the Moon and planets, and occultations.

- **Period** — 7, 30, 90 or 180 days, 1, 5 or 10 years.
- **Direction** — upcoming, past, or both.
- **Include occultations** — the Moon passing in front of a star, with immersion or emersion, dark or bright limb, and the star's magnitude.
- **Include ISS / Starlink passes** — adds satellite passes to the list.
- **Eclipses only** — narrows it to eclipses, with local circumstances (magnitude and times for your site).

- Tap a row and **the chart jumps to that moment**, so you can see for yourself what the sky will look like.
- ⟳ recalculates. The first 5- or 10-year scan takes a few minutes; the result is cached and appears instantly thereafter.

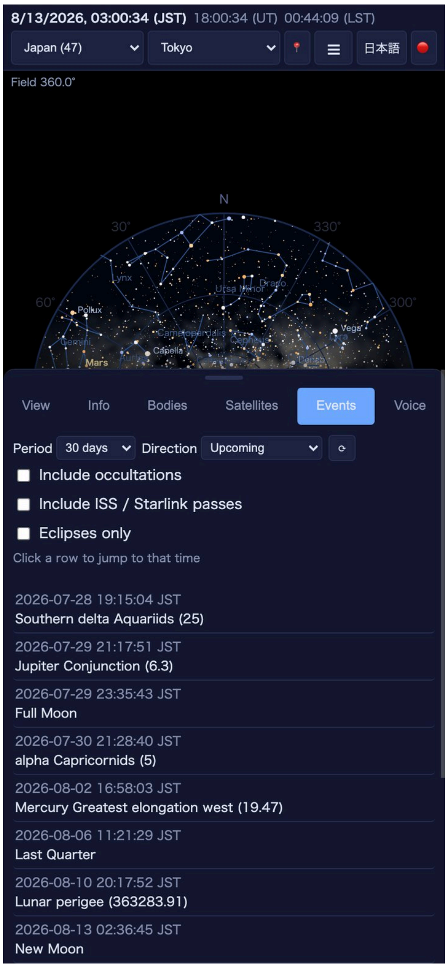

Figure 13. The Events tab

11. The Info tab

- **The sky for the date and place you have chosen** — sunset and sunrise, the beginning and end of twilight, the dark window, moonrise, moonset and lunar age, the transits of the Sun and Moon, the traditional lunisolar date and solar term, and local sidereal time. The heading names them both — "Tokyo — 22 August 2026" — and moving the clock to another date changes the table with it.
- **Selected object** — the position and rise/set times of whatever you last tapped on the chart.
- **Enter coordinates manually** — as in chapter 3.
- **Credits and sources** — every catalogue, ephemeris, image and library the app uses, with its source, its licence and the date it was last checked. It follows the language setting.

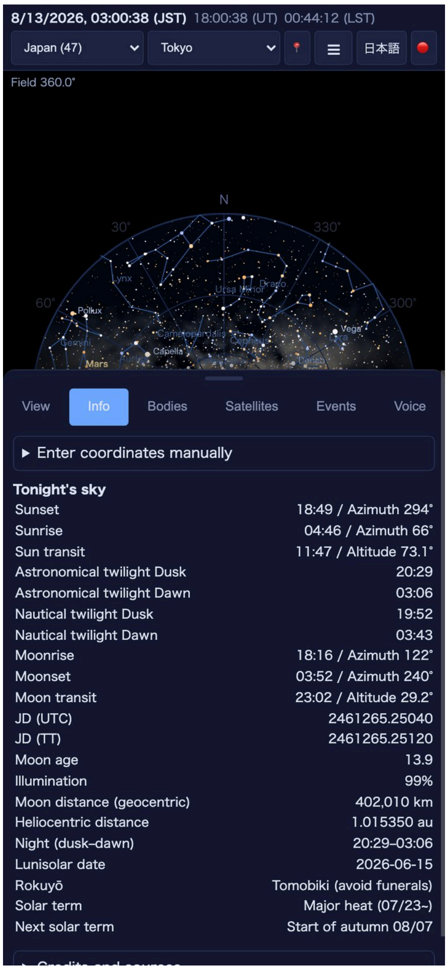

Figure 14. The Info tab, “Tonight’s sky”

12. The Voice tab

Astrarium can read the sky out loud, in Japanese or English independently of the interface language.

- **Speak tonight’s highlights** — what is worth looking at tonight.
- **Speak selected object** — the position and rise/set times of the object you picked on the chart.
- **Rate** and **Volume** — sliders.
- **Stop speaking** — interrupts it.

This is for the dark, when you would rather not look at a screen at all, or when your eye is at the eyepiece.

13. If something looks wrong

Location does not work

Settings → Privacy & Security → Location Services, and set Astrarium to “While Using the App”. You can also skip location altogether and type coordinates into the Info tab.

No satellites appear

Either the TLEs have never been downloaded, or playback is faster than ×600. With “Sunlit only” ticked, satellites in the Earth’s shadow are hidden.

TLEs will not download, or the address has been blocked

CelesTrak answers a caller that fetches the same group more often than every two hours with an error, and blocks the **IP address** if it continues. The block is against the connection, not against your device or Apple ID, so:

1. Stop fetching and leave it alone for a while. Astrarium does not touch the network within two hours of a fetch, so ordinary use will not get you blocked.
2. Switching networks (Wi-Fi to mobile data, or the other way) changes the address and often works. On shared Wi-Fi the cause may be someone else on the same connection.
3. If it does not clear, ask CelesTrak (celestrak.org) to lift it.

While blocked, **the elements you have already fetched still work**. They lose accuracy over a few days, but the chart and the pass predictions keep running without a connection.

No comets appear

The Comets button shows nothing when no comet is brighter than the limit, and fetching current elements needs a connection.

No meteor rate is shown

You are more than a day from the maximum, or outside the activity period. The rate is also zero in daylight.

Too few or too many stars

Move the magnitude-limit slider. To match a real sky, a city is about 4, the suburbs 5-something, and a dark mountain site 6.5.

The event list is slow

Only the first 5- or 10-year scan, which takes a few minutes. It is cached afterwards.

14. Glossary

Term	Meaning
Magnitude	Brightness; smaller is brighter, and one magnitude is a factor of about 2.5
Azimuth / altitude	Direction round the horizon (north 0°, east 90°) and angle above it
RA / Dec	Coordinates fixed to the sky — the system star catalogues use
Local sidereal time	The right ascension currently on the meridian
Universal Time	Time at longitude 0°
Moon age	Days since new Moon; 0 is new, about 14.8 is full
Illumination	The fraction of the Moon's disc that is lit
Airmass	Thickness of atmosphere, 1 at the zenith; the larger, the worse the seeing
ZHR	Meteors per hour with the radiant overhead and a 6.5-magnitude sky
Radiant	The point the meteors of a shower appear to come from
TLE	A satellite's orbital elements; they go stale in a few days and need refreshing

15. Credits and contact

Every source Astrarium computes from is listed in the app under **Info** → **Credits and sources**. The principal ones are:

- Planetary ephemeris: JPL DE421
- Stars: Yale Bright Star Catalogue, Tycho-2
- Deep sky: the Messier and Caldwell catalogues; photographs from Wikimedia Commons (freely licensed images only)
- Satellites: Celestrak (TLEs), SGP4/SDP4
- Comets and asteroids: Minor Planet Center, JPL Small-Body Database

- Earth orientation: IERS finals2000A

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